

This document details several "Hands-on Exercises" related to chemical and generic data analysis using data modeling and clustering tools, likely in a Pipeline Pilot environment.

Combined Document: Analytics Training Exercises

1. Example of ML Model Times 1

This protocol serves as an example environment for comparing various R-based Machine Learning models.

Models Included:

- **Learn R Linear Discriminant Analysis Model (LDA):** A classification model for a categorical property. Uses molecular properties like **ALogP**, **Molecular_Weight**, and **Num_H_Donors** as features. 2222222
 - **Learn R Logistic Regression Model:** Builds a logistic regression model for binary or multiclass classification. Uses features like **ALogP**, **Molecular_Weight**, **Num_RotatableBonds**, and **ECFP_4**. 3333333
 - **Learn R XGBoost Model:** Builds an extreme gradient boosting model for continuous or categorical properties. Key parameters for tuning include **Max Trees** (default 100), **Learning Rate** (default 0.3), **Max Depth** (default 7), **Data Fraction** (default 1.0), and **Descriptor Fraction** (default 0.7). 4444444444444444444444444444
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2. Hands-on Exercise 1: Analyse Auto Makes 5

Objective: Calculate the frequency of different car makes and their price statistics from the imports-85.txt dataset.

Key Steps:

1. **Read Data:** Use **Delimited Text Reader** on data\Tables\imports-85.txt6.
 2. **Calculate Frequencies:** Use **Generate Frequencies** on the make property. The output is collected **At End** and sorted by **Frequency** in **Descending** order7777777777.
 3. **Calculate Basic Statistics:** Use **Basic Statistics by Category** to calculate the **Mean**, **StdDevSample**, and **N** (count) of the price property, categorized by the make property8888888888.
 4. **Sort and View:** Sort the resulting statistics by the **Mean** of price in **Ascending** order and display it using the **HTML Table Viewer**9999.
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3. Hands-on Exercise 2a: Clustering Estrogen Agonists with Cluster Molecules 10

Objective: Cluster chemical compounds (Estrogen Antagonists) using a relocation-based clustering algorithm.

Key Steps:

1. **Read Data:** Use **SD Reader** to read data\Examples\EstrogenAntagonists.sd¹¹.
 2. **Clustering:** Use the **Cluster Molecules** component.
 - Clustering is based on distances calculated from the **FCFP_4** fingerprint (a predefined property set)¹².
 - The target cluster size is set by **AvgNumberPerCluster** (default 20)¹³¹³¹³.
 - The distance metric used is **Tanimoto** for fingerprints¹⁴.
 3. **Visualization:** Format the calculated **DistanceToClosest** property to 4 significant figures and display the cluster members and centers using the **HTML Molecular Cluster Viewer**¹⁵¹⁵¹⁵.
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4. Hands-on Exercise 2b: Clustering Animal Data using R AGNES ¹⁶

Objective: Cluster animal data using the hierarchical agglomerative nesting method (AGNES) in R.

Key Steps:

1. **Read Data:** Use **Delimited Text Reader** to read data\Tables\zoo.txt¹⁷.
 2. **Clustering:** Use **R Cluster Agnes** component.
 - Use **AllPropertiesOnFirstData** for distance calculations, ignoring animalname¹⁸¹⁸¹⁸.
 - **DistanceMetric** is set to **ManhattanBlock**¹⁹.
 - **NormalizeData** is set to **False**²⁰.
 - **LabelsProperty** is set to animalname for plot labels²¹.
 3. **Visualization:** Output results as cluster assignments, and display a cluster plot and dendrogram using the **R Plot Viewer**²²²²²²²².
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5. Hands-on Exercise 2c: Clustering Chemistry data using R AGNES ²³

Objective: Cluster Estrogen Antagonists using R AGNES, similar to 2b but with chemical data and a small test for the impact of fixed length.

Key Steps:

1. **Read Data:** Use **SD Reader** to read data\Examples\EstrogenAntagonists.sd²⁴.
 2. **Clustering:** Use **R Cluster Agnes** component.
 - Property to cluster on: **FCFP_4**²⁵²⁵.
 - **DistanceMetric:** **Euclidean**.
 - **FixedLength:** Set to **128** for the primary run²⁶²⁶²⁶²⁶.
 - The exercise suggests comparing this with a **FixedLength** of **64**²⁷.
 3. **Visualization:** Display results using the **Html Molecular Cluster Viewer** and the **R Plot Viewer**²⁸.
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6. Hands-on Exercise 3: Three Different Models of Auto Price 29

Objective: Compare performance of Least Squares (LS), Partial Least Squares (PLS), and k-Nearest Neighbor (kNN) models for predicting car **price** from auto imports data.

Key Steps:

1. **Data Preparation:** Use **Delimited Text Reader** on imports-85.txt. Select a broad set of properties (wheelbase, length, width, height, weight, etc.) for modeling³⁰.
 2. **Data Split:** Use a **Random Percent Filter** to assign 70% of data for training and 30% for testing³¹.
 3. **Model Building:** Build three models using the **Learn Property** component:
 - **Least Squares (LS)**³².
 - **PLS** with **NumberOfComponents** = 4333333.
 - **kNN** with **Number of Nearest Neighbors** = 5 and **Number of XV Groups** = 534.
 4. **Analysis:** The goal is to compare which model best fits the training data and gives the best test set predictions³⁵.
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7. Hands-on Exercise 4b: Two R LM Models for Auto Price 36

Objective: Build and evaluate a specific Linear Model (LM) in R for car price prediction.

Key Steps:

1. **Read Data:** Use **Delimited Text Reader** on imports-85.txt³⁷.
 2. **Model Building:** Use **Learn R Linear Model** to predict **price** as a function of height, weight, horsepower, and citympg³⁸.
 3. **Evaluation:**
 - View the **FitSummary** using **Notepad Viewer** to find R-squared and prediction R-squared values for overfitting assessment³⁹.
 - View the **FitPlot** using **R Plot Viewer** to inspect model results, specifically the residuals plot, and consider potential term drops or variable transformations (e.g., transforming **price**)⁴⁰.
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8. Hands-on Exercise 5 for Chemists: Bayesian Model for MAO Activity 4141

Objective: Develop a Bayesian model to predict Monoamine Oxidase (MAO) activity.

Key Steps:

1. **Read Data:** Read Chemistry Data / MAO / maoAct.sd using **SD Reader**⁴²⁴².
2. **Define Classes:** Use **PilotScript** (Custom Manipulator) to create a categorical property Class:
 - Class := 'active' if activity >= 2
 - Class := 'inactive' otherwise⁴³⁴³⁴³⁴³.

3. **Frequency Analysis:** Use **Generate Frequencies** on the activity property to count highly, moderately, and weakly active compounds44444444.
 4. **Model Building:** Use **Learn Good Molecules** (Bayesian classifier) to distinguish high actives (activity == 3) from all others. Uses default **PredefinedSet** properties and includes **Validate Models** and **Equipopulate Bins** options4545454545454545454545454545454545.
 5. **Application:** The exercise instructs finding the 10 most MAO inhibitor-like compounds and identifying possible false negatives46.
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9. Hands-on Exercise 6 for All & 6 for Chemists (Combined Model Tree Analysis)

These exercises focus on using **Recursive Partitioning (RP) Tree Models** and **Cross-validated RP Tree Models** to analyze the quality of the tree models (overfitting/generalization).

Common Tools and Concepts:

- **Learners:** Learn RP Tree Model and Learn Cross-validated RP Tree Model.
 - **Output:** The exercises emphasize determining the "**best**" **pruned tree** by comparing training data metrics (from Learn RP Tree Model) against **cross-validated ROC scores** (from Learn Cross-validated RP Tree Model)47474747.
 - **Tree Options:** Key parameters adjusted include **Minimum Samples Per Node** (set to 3 or 5 in the examples)48484848.
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10. Hands-on Exercise 4b (Linear Regression Example, Transformed)

Note: This is a continuation/alternative to Exercise 4b's base instruction found in other files, likely focusing on a log-transformed response for improved fitting.

The instruction in the introductory document suggested transforming **price** to get a better fit49, implying a secondary exercise not explicitly detailed in the main parameters provided here, but is part of the common tutorial flow.

I understand you want the combined document from the uploaded .ppxml files presented in a clean document format, not XML.

Here is the compiled and organized content from your "Hands-on Exercise" protocols:

Combined Document: Advanced Analytics and Modeling Exercises

1. Exercise Overview and ML Model Examples (Files: Example of ML model times.ppxml, Hands-on Exercise 1.ppxml, Hands-on Exercise 3.ppxml)

Exercise	Objective	Data & Context	Key Component s/Model Types	
ML Model Times	Example environment for comparing R-based Machine Learning models ¹ .	Uses molecular properties like ALogP , Molecular_Weight , and Num_H_Donors as features ² .		Learn R Linear Discriminant Analysis Model (LDA) , Learn R Logistic Regression Model , and Learn R XGBoost Model ³ .
Hands-on Exercise 1	Calculate frequency and price statistics of car makes ⁴ .	Data source is data\Tables\imports-85.txt ⁵ .		Delimited Text Reader , Generate Frequencies (on make), Basic Statistics by Category (on price by make) ⁶ .
Hands-on Exercise 3	Compare three regression models for predicting car price ⁷ .	Data source is imports-85.txt ⁸ . Training set (70%), Test set (30%) split using Random Percent Filter ⁹ .		Learn Property using Least Squares (LS) , Partial Least Squares (PLS) , and k-Nearest Neighbor (kNN) models ¹⁰ .

2. Clustering Exercises (Files: Hands-on Exercise 2a.ppxml, Hands-on Exercise 2b.ppxml, Hands-on Exercise 2c.ppxml)

Exercise	Objective	Data & Method	Visualization / Output	
Hands-on Exercise 2a	Cluster Estrogen Antagonists ¹¹ using relocation-based clustering ¹² .		Data: EstrogenAntagonists.sd 13. Method: Cluster Molecules using FCFP_4 fingerprint and Tanimoto distance ¹⁴ .	HTML Molecular Cluster Viewer ¹⁵ .
Hands-on Exercise 2b	Cluster animal data ¹⁶ using hierarchical agglomerative nesting in R ¹⁷ .		Data: zoo.txt ¹⁸ . Method: R Cluster Agnes using ManhattanBlock metric ¹⁹ .	R Plot Viewer (for cluster plot and dendrogram) ²⁰ .
Hands-on Exercise 2c	Cluster chemical data (EstrogenAntagonists.sd) ²¹ using R AGNES with varying fingerprint length ²² .		Data: EstrogenAntagonists.sd 23. Method: R Cluster Agnes using FCFP_4 with FixedLength set to 128 (suggested comparison with 64) ²⁴ .	HTML Molecular Cluster Viewer and R Plot Viewer ²⁵ .

3. Classification and Feature Importance (Files: Hands-on Exercise 5 for All.ppxml, Hands-on Exercise 7 for All.ppxml, Hands-on Exercise 7 for Chemists.ppxml, Hands-on Exercise 9 for Chemists.ppxml)

Exercise	Model / Goal	Training & Output Details	Key Insight	
Hands-on Exercise 5 for	Build a Bayesian Classifier for "Toyotanness" ²⁶		Model: Toyotanness (Bayesian/SciTegic) ²⁷ . Evaluation:	The training data shows XV ROC AUC of 0.987 , suggesting

Exercise	Model / Goal	Training & Output Details	Key Insight	
All	.		Leave-one-out Cross-Validation is used ²⁸ . Output: ROC Plot Viewer to compare performance ²⁹ .	excellent predictive power for "Toyotanness" ³⁰ .
Hands-on Exercise 7 for All	Build RP Forest Models for classifying neurological vs. autism abstracts ³¹ .		Feature Engineering: Simple String Fingerprint (SFP) is used ³² . Manipulation: A custom script overwrites common, obvious words (autism, neurological) with random values (rand()) to force the model to identify less obvious differentiating keywords ³³ .	A deliberate step to study the importance of less obvious textual features ³⁴ .
Hands-on Exercise 7 for Chemists	Build RP Forest Models for classifying hERG channel blockage (class) ³⁵ .		Models to Compare: A default Random Forest (500 trees) ³⁶ vs. a Bagged Forest (10 trees, all descriptors considered) ³⁷ .	The prompt asks which model should yield more accurate predictions on new data (often the random forest over the bagged forest

Exercise	Model / Goal	Training & Output Details	Key Insight	
				when hyperparameters are optimized, but here, the core exercise is to compare them) ³⁸ .
Hands-on Exercise 9 for Chemists	Build a Cross-validated RP Forest Regression Model for Plasma Protein Binding (PPB) (PBexp(%)) ³⁹ .		Model Creation: Uses a custom learner subprotocol ⁴⁰ . Feature Selection: Predicts PBexp(%) from PredefinedSet properties ⁴¹ . Model Output: Set to Report with Variable Importance estimated, and Min Fingerprint Feature Count set to 64 ⁴² .	Focus is on analyzing the reported variable importance measures to determine which properties influence PPB ⁴³ .

4. Advanced Optimization (Files: Hands-on Exercise 11 for All.ppxml, Hands-on Exercise 11 for Chemists.ppxml, Hands-on Exercise 13 Challenge.ppxml, Hands-on Exercise 13 Custom Scoring Protocol.ppxml)

Exercise	Tool	Objectives & Goal	Key Settings	
Hands-on Exercise 11 for All	Pareto Subset Optimizer	Design an optimal Stock Portfolio ⁴⁴ . Goal 1: Maximize Mean Return (PctReturn_Mean) ⁴⁵ . Goal 2: Minimize Mean Variance (PctChange_Variance_Mean) (Risk) ⁴⁶ .		Subset Size: 2047. What to Output: Grouped Subset Members ⁴⁸ .
Hands-on Exercise 11 for Chemists	Pareto Subset Optimizer	Design a diverse, drug-like Chemical Library ⁴⁹ . Goal 1: Maximize Mean of druglike property ⁵⁰ . Goal 2: Maximize Diversity (FCFP_4_UniqueValueCount) ⁵¹ .		Subset Size: 5052. What to Output: Grouped Subset Members ⁵³ .
Hands-on Exercise 13 Challenge	Pareto Subset Optimizer (Advanced/Custom)	Design a set of maximally dissimilar compounds for an HTS campaign ⁵⁴ . Goal 1 (Custom): Minimize the calculated maximum similarity (ClosestSimilarities_Max) ⁵⁵ . Goal 2: Maximize a molecular property like druglike ⁵⁶ .		Subset Size: 9657. Number of Subsets: 4058. Number of Iterations: 10059. Uses a custom scoring protocol ⁶⁰ .
13 Custom	Sub-protocol	Computes the single numerical "score"		Steps: 1. Molecular Similarity NxN

Exercise	Tool	Objectives & Goal	Key Settings
Scoring Protocol	used by Exercise 13 Challenge	(Max) that the Pareto optimizer minimizes ⁶¹ .	(FCFP_4, Tanimoto) ⁶² . 2. Custom Manipulator (PilotScript) to removevalue(Closest Similarities, 1) to eliminate self-similarity (which is always 1.0) ⁶³ . 3. Basic Statistics to find the Max similarity value among the remaining neighbors ⁶⁴ .